



Two classes of spectral three-term derivative-free method for solving nonlinear equations with application

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Abstract

Solving large-scale systems of nonlinear equations (SoNE) is a central task in mathematics that traverses different areas of applications. There are several derivative-free methods for finding SoNE solutions. However, most of the methods contributed to find SoNE solutions involve a monotone cost function. Methods dealing with pseudomonotone cost function remain rare. In this paper, we introduce two classes of derivative-free spectral three-term methods to solve large-scale continuous pseudomonotone SoNE. We combine the projection method of Solodov and Svaiter with the structure of the recently developed spectral three-term conjugate gradient method for unconstrained optimization by Amini and Faramarzi. We prove that the proposed methods possess sufficient descent property, trust region property, and global convergence without relying on Lipschitz continuity. Numerical experiments show that the proposed methods are efficient and competitive with existing methods. Finally, the proposed methods have been successfully applied to recover a sparse signal from incomplete and contaminated sampling measurements, yielding promising results.

Keywords Iterative method · Nonlinear equations · Large-scale systems · Derivative-free method · Projection method

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1 Introduction

Given a nonempty, closed, and convex set $\mathcal{C}$ in the n -dimensional Euclidean space $\mathbb{R}^n$, the problem we consider in this article is to find a vector $x \in \mathcal{C}$ satisfying

$$F(x) = 0, \quad (1)$$

where $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuous mapping. Various methods have been proposed to solve (1). Among them, Newton's method and quasi-Newton methods [1–7] are highly regarded due to their local superlinear convergence property. However, these methods are not usually preferred for large-scale nonlinear systems of equations, since they require solving a linear system using either the Jacobian matrix or an approximation of it, which can be computationally expensive.

The simplicity and low storage requirements of certain first-order optimization methods, such as spectral gradient methods, conjugate gradient methods, and spectral conjugate gradient methods, have made them a popular choice for addressing large-scale unconstrained optimization problems. As a result, researchers have extended these methods to handle large-scale systems of nonlinear equations [8–11]. It is worth mentioning that the projection technique presented by Solodov and Svaiter [12] is an essential component in extending most of the first-order optimization methods to solve (1).

In recent years, the conjugate gradient method has been improved by introducing two important generalizations: the three-term conjugate gradient methods and the spectral conjugate gradient methods. These generalizations have improved the efficiency of the conjugate gradient method in terms of computational results and convergence properties [13]. The first three-term conjugate gradient method was proposed by Beale [14], and since then, extensive research has been conducted on these methods, as seen in [15–20]. The spectral conjugate gradient method was first proposed by Birgin and Martinez [21]. The method combines the technique of conjugate gradient and spectral gradient methods to improve the efficiency of conjugate gradient methods. Researchers have built upon this idea by developing spectral conjugate gradient methods [22–24].

On the other hand, the convergence properties of mathematical algorithms have gained significant interest in recent times owing to their crucial role in the approximation of solutions to real-world problems. Integrating inertial techniques is an empirical approach to expedite an optimization method, and their use dates back to the heavy ball method [25]. The technique has attracted the interest of researchers [26] and has found widespread application in convex and convex composite optimization problems, particularly when the objective function is smooth, see, e.g. [27–30]. Furthermore, they have been extended to smooth nonconvex problems, as seen in [31].

Inspired by the inertial technique, Abubakar et al. [32] proposed an inertial derivative-free method for solving SoNE. The global convergence of their proposed method depends on a summability condition that constrains the selection of the inertial extrapolation step length. Around the same time, Ibrahim et al. [33] presented a method that combined the idea of the inertial technique with the hyperplane projection method and the structure of the conjugate gradient method to solve (1). The global convergence of the method proposed by Ibrahim et al. [33] adeptly chooses the

inertial step length in a manner that circumvents the need for the summability condition stipulated in [32]. Following the work of Ibrahim et al. [33], researchers have proposed derivative-free projection methods with inertial step to solve (1) [34–41]. The common feature of these methods is that they are stable in descent property and convergence, and the computing performance is satisfactory. However, the global convergence of these methods was established under the assumption that the underlying mapping is monotone and Lipschitz continuous. To weaken these assumptions, Jian et al. [34] proposed a derivative-free inertia method for solving (1) and proved the global convergence under the assumption that the underlying mapping is continuous and pseudomonotone.

Quite recently, Li et al. [42] introduced a spectral conjugate gradient method that satisfies the sufficient descent property, which is effective in computational performance. However, it is important to note that this method's convergence property has only been proven for uniformly convex functions. To address this limitation, Amini and Faramarzi [13] presented a modified version of the spectral three-term CG method introduced by Li et al. [42]. This modification was aimed at improving the global convergence of the method for general nonlinear functions. Additionally, they introduced a new spectral parameter based on the strong convergence properties of quasi-Newton methods.

In this paper, using the projection method proposed by Solodov and Svaiter [12], we design two classes of derivative-free spectral three-term methods to solve (1). The proposed methods use the Brazilian and Bowerin-like spectral parameter proposed by Abubakar et al. [43] and the structure of the conjugate gradient method developed by Amini and Faramarzi [13]. Under a weaker monotonicity condition and without relying on Lipschitz continuity, the proposed methods are established to be globally convergent. Furthermore, the proposed methods are shown to possess sufficient descent and trust region property. Finally, several numerical experiments are performed to support the obtained theoretical results.

The structure of this paper is organized as follows. The next section provides an overview of the preliminary knowledge and introduces several helpful lemmas. In the third section, we present the algorithms for the proposed methods and establish their global convergence property. Section 4 presents the numerical implementation of the proposed methods in solving nonlinear equations and signal reconstruction problem. Finally, the last section concludes the paper by summarizing the findings and contributions.

2 Preliminaries

In this section, we recall some definitions and properties that will be used in the paper.

Definition 2.1 An operator $F : \mathcal{C} \rightarrow \mathcal{C}$ is said to be

- (i) **monotone** on $\mathcal{C}$ if

$$(F(x) - F(y))^T(x - y) \geq 0, \quad \forall x, y \in \mathcal{C};$$

(ii) **pseudomonotone** on $\mathcal{C}$, in the sense of Karamardian [44], if

$$F(x)^T(y - x) \geq 0 \implies F(y)^T(y - x) \geq 0, \quad \forall x, y \in \mathcal{C}.$$

Definition 2.2 A **projection** $P_{\mathcal{C}} : \mathbb{R}^n \rightarrow \mathcal{C}$ is defined by

$$P_{\mathcal{C}}(x) = \arg \min \{\|y - x\| : y \in \mathcal{C}\}, \quad \forall x \in \mathbb{R}^n.$$

Some well-known properties of the projection operator are stated in the following result.

Lemma 2.3 [45] *Let $\mathcal{C} \subseteq \mathbb{R}^n$ be a nonempty, closed and convex set. Then, the following statements hold*

- (i) $(x - P_{\mathcal{C}}(x))^T(P_{\mathcal{C}}(x) - y) \geq 0, \quad \forall x \in \mathbb{R}^n, \quad \forall y \in \mathcal{C}.$
- (ii) $\|P_{\mathcal{C}}(x) - P_{\mathcal{C}}(y)\| \leq \|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$
- (iii) $\|P_{\mathcal{C}}(x) - y\|^2 \leq \|x - y\|^2 - \|x - P_{\mathcal{C}}(x)\|^2, \quad \forall x \in \mathbb{R}^n, \quad y \in \mathcal{C}.$

Lemma 2.4 [35] *Let $\{x_k\}$ and $\{z_k\}$ be two sequences of nonnegative real numbers such that $\sum_{k=0}^{\infty} z_k < \infty$. If $x_{k+1} \leq x_k + z_k$ for all $k \geq 0$, then the sequence $\{x_k\}$ is convergent.*

3 Algorithms and convergence analysis

Let Γ denote set of solutions to (1). Throughout this paper, we assume the following

- (A1) Γ is nonempty.
- (A2) F is pseudomonotone.

Next, we give the recently proposed conjugate gradient method of Amini and Faramarzi [13] for unconstrained optimization problem

$$\min f(x), \quad x \in \mathbb{R}^n \quad (2)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable and its gradient $g(x)$ is available. For the sake of simplicity, the gradient of f at point x_k is denoted by g_k . The Amini and Faramarzi [13] conjugate gradient method generates a sequence of iterates $\{x_k\}$ given by

$$x_{k+1} = x_k + \tau_k d_k, \quad k \geq 0, \quad (3)$$

where x_k is the current iterate, τ_k is called the step length which is determined by some line search and d_k is the search direction defined by

$$d_0 = -g_0, \quad d_k = -\theta_k g_k + \beta_k^N d_{k-1} - \tilde{v}_k y_{k-1} \quad \forall k > 0 \quad (4)$$

where

$$\beta_k^N = \frac{g_k^T y_{k-1}}{w_k}, \quad \tilde{v}_k = \frac{g_k^T d_{k-1}}{w_k} \quad (5)$$

$$w_k = \max \left\{ \psi \|d_{k-1}\| \|y_{k-1}\|, \quad d_{k-1}^T y_{k-1}, \quad \|g_{k-1}\|^2 \right\}, \quad (6)$$

in which $y_{k-1} = g_k - g_{k-1}$, $\psi > 0$ and θ_k is a spectral parameter computed based on quasi-Newton method. Interested readers can refer to [13] and references therein for more details. We extend the idea of the conjugate gradient method of Amini and Faramarzi [13] to solve the systems of nonlinear (1). In the next subsections, we will present two new methods, namely **spectral three-term derivative-free projection method** and **inertia spectral three-term derivative-free projection method**, referred to as **STTDFPM** and **ISTTDFPM** respectively.

3.1 STTDFPM algorithm and global convergence

In this subsection, based on the formulae (3)-(6), we construct a derivative-free projection method for solving (1). For convenience, we define the projection operator $P_{[a,b]} : \rightarrow [a, b]$, which projects onto the closed and bounded interval $[a, b]$, as

$$P_{[a,b]}(x) = \max \{a, \min\{b, x\}\}, \quad \forall x \in \mathbb{R}. \quad (7)$$

So given the parameters $0 < \alpha_{\min} < \alpha_{\max}$, $r \in (0, 1)$, $0 < \eta_1 \leq \eta_2$, $\sigma \in (0, 1)$, and $\psi > 0$, our approach is a projection procedure that generates a sequence $\{z_k\}$ such that $z_k = x_k + \tau_k d_k$, where the step length τ_k is obtained as $\tau_k = \beta^{i_k}$ with i_k the least nonnegative integer such that

$$-F(z_k)^T d_k \geq \sigma \tau_k P_{[\eta_1, \eta_2]}(\|F(z_k)\|) \|d_k\|^2 \quad (8)$$

and d_k is the line search direction given by

$$d_k = \begin{cases} -F(x_0), & \text{if } k = 0, \\ -\alpha_k^I F(x_k) + \beta_k d_{k-1} - \alpha_k^{II} y_{k-1} & \text{if } k \geq 1 \end{cases} \quad (9)$$

where

$$\begin{aligned} y_{k-1} &= F(x_k) - F(x_{k-1}), & s_{k-1} &= x_k - x_{k-1} + r y_{k-1}, \\ \alpha_k^I &= P_{[\alpha_{\min}, \alpha_{\max}]} \left(\frac{s_{k-1}^T y_{k-1}}{y_{k-1}^T y_{k-1}} \right), & \beta_k &= \frac{F(x_k)^T y_{k-1}}{v_k}, & \alpha_k^{II} &= \frac{F(x_k)^T d_{k-1}}{v_k} \end{aligned} \quad (10)$$

with

$$v_k = \max \left\{ \psi \|d_{k-1}\| \|y_{k-1}\|, \quad \|F(x_{k-1})\|^2 \right\}. \quad (11)$$

Remark 3.1 The search direction outlined in (4)–(6) is primarily tailored for addressing unconstrained optimization problems. On the other hand, the derivative-free search direction outlined in (9)–(11) is a modified version of the former, specifically designed for tackling systems of nonlinear equations. Importantly, the distinct advantage of the search direction (9)–(11) over its counterpart (4)–(6), is its applicability to solving nonsmooth equations.

Remark 3.2 The adaptive line search (8) was proposed in [46] and it was shown to be well defined. The parameter η_2 was proposed to keep the right-hand side of (8) from growing too large. The parameter η_1 was selected to avoid the case where $F(z_k) = 0$ and $z_k \notin \mathcal{C}$. It should be also noted that the spectral parameter α_k^I originated in [43].

Remark 3.3 In (11), v_k is positive and therefore α_k^{II} and β_k are well-defined.

Remark 3.4 For all k , the spectral parameter α_k^I is uniformly bounded. That is, $\alpha_{\min} \leq \alpha_k^I \leq \alpha_{\max}$.

The new algorithm **STTDFPM** can be now outlined as follows.

Algorithm 1 Spectral Three-Term Derivative-Free Projection Method (STTDFPM).

Input : A function F , a projection P on the set $\mathcal{C}$, an initial point $x_0 \in \mathcal{C}$, and a stopping tolerance $\epsilon > 0$

Output : A solution x^* to (1)

Parameters : $0 < \alpha_{\min} < \alpha_{\max}$, $r \in (0, 1)$, $\beta \in (0, 1)$, $\sigma \in (0, 1)$, $\gamma \in (0, 2)$, $\psi > 0$, $\mu > 0$

Initialization: $k \leftarrow 0$, $z_k \leftarrow 0$, $\mu_k \leftarrow 0$

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1 while  $\|F(x_k)\| > \epsilon$  do
2   find  $d_k$  according to (9)
3   if  $\|d_k\| \approx 0$  then
4      $x^* \leftarrow x_k$ 
5     break
6   end
7   find  $\tau_k$  according to (8)
8    $z_k \leftarrow x_k + \tau_k d_k$ 
9   if  $z_k \in \mathcal{C}$  and  $\|F(z_k)\| \leq \epsilon$  then
10     $x^* \leftarrow z_k$ 
11    break
12  end
13   $\mu_k \leftarrow \frac{F(z_k)^T (x_k - z_k)}{\|F(z_k)\|^2}$ 
14   $x_{k+1} \leftarrow P_{\mathcal{C}}(x_k - \gamma \mu_k F(z_k))$ 
15   $k \leftarrow k + 1$ 
16   $x^* \leftarrow x_k$ 
17 end

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Lemma 3.5 *The search direction d_k generated by Algorithm 1 satisfies the sufficient descent condition*

$$F(x_k)^T d_k \leq -\alpha_{\min} \|F(x_k)\|^2, \quad (12)$$

and the trust region property

$$\alpha_{\min} \|F(x_k)\| \leq \|d_k\| \leq \left(\alpha_{\max} + \frac{2}{\psi} \right) \|F(x_k)\|. \quad (13)$$

Proof For $k = 0$, it is not difficult to see that the search direction d_k satisfies (12). Noting Remark 3.4, for $k > 0$, we have

$$\begin{aligned} F(x_k)^T d_k &= -\alpha_k^I \|F(x_k)\|^2 + \beta_k F(x_k)^T d_{k-1} - \alpha_k^{II} F(x_k)^T y_{k-1} \\ &= -\alpha_k^I \|F(x_k)\|^2 + \frac{F(x_k)^T (y_{k-1})}{v_k} F(x_k)^T d_{k-1} - \frac{F(x_k)^T d_{k-1}}{v_k} F(x_k)^T (y_{k-1}) \\ &= -\alpha_k^I \|F(x_k)\|^2 \\ &\leq -\alpha_{\min} \|F(x_k)\|^2. \end{aligned}$$

Meanwhile, according to $\max\{\psi \|d_{k-1}\| \|y_{k-1}\|, \|F(x_{k-1})\|^2\} \geq \psi \|d_{k-1}\| \|y_{k-1}\|$, then

$$\begin{aligned} \|d_k\| &= \| -\alpha_k^I F(x_k) + \beta_k d_{k-1} - \alpha_k^{II} y_{k-1} \| \\ &\leq |\alpha_k^I| \|F(x_k)\| + |\beta_k| \|d_{k-1}\| + |\alpha_k^{II}| \|y_{k-1}\| \\ &\leq \alpha_{\max} \|F(x_k)\| + \left| \frac{F(x_k)^T y_{k-1}}{v_k} \right| \|d_{k-1}\| + \left| \frac{F(x_k)^T d_{k-1}}{v_k} \right| \|y_{k-1}\| \\ &\leq \alpha_{\max} \|F(x_k)\| + \frac{\|F(x_k)\| \|y_{k-1}\|}{\psi \|d_{k-1}\| \|y_{k-1}\|} \|d_{k-1}\| + \frac{\|F(x_k)\| \|d_{k-1}\|}{\psi \|d_{k-1}\| \|y_{k-1}\|} \|y_{k-1}\| \\ &= \left(\alpha_{\max} + \frac{2}{\psi} \right) \|F(x_k)\|. \end{aligned} \quad (14)$$

Combining with (12) and Cauchy Schwartz inequality, the proof follows. $\square$

Lemma 3.6 *Suppose that assumptions (A1) and (A2) hold and the sequence $\{x_k\}$ is generated by Algorithm 1. For any $x^* \in \Gamma$, the sequence $\{\|x_k - x^*\|\}$ is convergent. Furthermore, the sequence $\{x_k\}$ is bounded and*

$$\lim_{k \rightarrow \infty} \tau_k \|d_k\| = 0. \quad (15)$$

Proof From Remark 3.2, it is clear that the parameter $\mu_k = \frac{F(z_k)^T (x_k - z_k)}{\|F(z_k)\|^2}$ (see line 13 of Algorithm 1) is well-defined. Furthermore, from the line search (8), we have

$$F(z_k)^T (x_k - z_k) = -\tau_k F(z_k)^T d_k \geq \sigma \tau_k^2 \|F(z_k)\| \|d_k\|^2. \quad (16)$$

Therefore, $\mu_k \geq 0$. By (A2), Lemma 2.3(iii), (14) and (16), it follows that

$$\begin{aligned}
 0 \leq \|x_{k+1} - x^*\|^2 &= \|P_{\mathcal{C}}(x_k - \gamma\mu_k F(z_k)) - P_{\mathcal{C}}(x^*)\|^2 \\
 &\leq \|x_k - \gamma\mu_k F(z_k) - x^*\|^2 \\
 &\leq \|x_k - x^*\|^2 - 2\gamma\mu_k F(z_k)^T(x_k - x^*) + \gamma^2\mu_k^2\|F(z_k)\|^2 \\
 &= \|x_k - x^*\|^2 - 2\gamma\mu_k F(z_k)^T(x_k - z_k) - 2\gamma\mu_k F(z_k)^T(z_k - x^*) \\
 &\quad + \gamma^2\mu_k^2\|F(z_k)\|^2 \\
 &\leq \|x_k - x^*\|^2 - 2\gamma\mu_k F(z_k)^T(x_k - z_k) + \gamma^2\mu_k^2\|F(z_k)\|^2 \\
 &\leq \|x_k - x^*\|^2 - \gamma(2 - \gamma)\sigma^2\|x_k - z_k\|^4. \tag{17}
 \end{aligned}$$

Hence, the sequence $\{\|x_k - x^*\|\}$ is both non-increasing bounded below and therefore convergent, which implies that $\{x_k\}$ is bounded.

Taking into account of (17) and taking summation for $k = 0, 1, 2, \dots$, we have

$$\gamma(2 - \gamma) \sum_{k=0}^{\infty} \sigma^2\|x_k - z_k\|^4 \leq \sum_{k=0}^{\infty} (\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2) < \infty.$$

This establishes the validity of (15), thereby concluding the proof. $\square$

Remark 3.7 By virtue of the continuity of the mapping F , we can ascertain the existence of a positive constant m_0 , such that for all k , $\|F(x_k)\| \leq m_0$. Therefore, from (13), it follows that $\{d_k\}$ is bounded.

Theorem 3.8 *The sequence $\{x_k\}$ generated by Algorithm 1 converges globally to some $x^* \in \Gamma$.*

Proof To establish that $\{x_k\}$ converges to the solution of (1), we will show that

$$\liminf_{k \rightarrow \infty} \|F(x_k)\| = 0. \tag{18}$$

Suppose that the equality (18) is not true. Then, there exist a $\epsilon > 0$ such that

$$\|F(x_k)\| \geq \epsilon, \quad \forall k. \tag{19}$$

This together with (12) implies that

$$\|d_k\| \geq \alpha_{\min}\|F(x_k)\| \geq \epsilon\alpha_{\min}, \quad \forall k. \tag{20}$$

From the above relation, it can easily be seen that $\tau_k \epsilon \alpha_{\min} \leq \tau_k \|d_k\|$. Therefore, by (13), it follows that $\lim_{k \rightarrow \infty} \tau_k = 0$. Now, for every $k \geq 0$, i_k is the smallest integer such that $\tau_k = \beta^{i_k}$ satisfies (8), therefore $\tilde{\tau}_k = \beta^{-1}\tau_k$ does not satisfy (8). In other words, if the sequence $\{\tau_k\}$ is satisfies (8), the sequence $\{\tilde{\tau}_k\}$ violates it. That is, for $\tilde{z}_k = x_k + \tilde{\tau}_k d_k$, we have

$$-F(\tilde{z}_k)^T d_k < \sigma \tilde{\tau}_k P_{[\eta_1, \eta_2]}(\|F(\tilde{z}_k)\|)\|d_k\|^2 \leq \eta_2 \sigma \tilde{\tau}_k \|d_k\|^2. \tag{21}$$

Given that $\{x_k\}$ and $\{d_k\}$ are bounded, we can choose subsequences that converge to their limit points. Let $\tilde{x}$ and $\tilde{d}$ denote the limit points of these corresponding subsequences. By allowing k to approach infinity in (21), we obtain

$$-F(\tilde{x})^T \tilde{d} \leq 0. \quad (22)$$

On the other hand, taking the limit of both sides of (12), we have

$$-F(\tilde{x})^T \tilde{d} \geq 0.$$

This is a contradiction and therefore we must have

$$\liminf_{k \rightarrow \infty} \|F(x_k)\| = 0.$$

Since the underlying mapping F is continuous, it implies the existence of an accumulation point $\tilde{x}$ in the sequence $\{x_k\}$ such that $F(\tilde{x}) = 0$, indicating that $\tilde{x} \in \Gamma$. According to (17), the sequence $\{\|x_k - \tilde{x}\|\}$ converges. Since $\tilde{x}$ is an accumulation point of $\{x_k\}$, it follows that $\{x_k\}$ converges to $\tilde{x}$. $\square$

3.2 ISTTDFPM algorithm and global convergence

In this subsection, the proposed algorithm is constructed using the projection method and the inertial extrapolation technique. The inertial extrapolation point u_k is calculated for each iteration k as a combination of the current iterate x_k and previous iterate x_{k-1} , namely,

$$u_k = x_k + t_k(x_k - x_{k-1}) \quad (23)$$

where

$$t_k = \begin{cases} \min \left\{ t, \frac{1}{k^2 \|x_k - x_{k-1}\|} \right\} & \text{if } x_k \neq x_{k-1} \\ t & \text{otherwise} \end{cases} \quad \forall k \geq 1. \quad (24)$$

Remark 3.9 Observe that by the definition of the inertial extrapolation step length (24), for all k , $0 \leq t_k \leq t < 1$. Moreover, it follows that for all k

$$t_k \|x_k - x_{k-1}\| \leq \frac{1}{k^2}, \quad (25)$$

which implies that

$$\sum_{k=0}^{\infty} t_k \|x_k - x_{k-1}\| < \infty. \quad (26)$$

Therefore, $\lim_{k \rightarrow \infty} t_k \|x_k - x_{k-1}\| = 0$. Thus, we can infer that

$$\lim_{k \rightarrow \infty} \|x_k - u_k\| = 0. \quad (27)$$

Algorithm 2 Inertia Spectral Three-Term Derivative-free Projection Method (ISTDFPM).

Input : A function F , a projection P on the set $\mathcal{C}$, an initial point $x_0 \in \mathcal{C}$, and a stopping tolerance $\epsilon > 0$

Output : A solution x^* to (1)

Parameters : $0 < \alpha_{\min} < \alpha_{\max}$, $r \in (0, 1)$, $\beta \in (0, 1)$, $\sigma \in (0, 1)$, $\gamma \in (0, 2)$, $\psi > 0$

Initialization: $x_1 \leftarrow x_0, u_1 \leftarrow x_0, k \leftarrow 1, z_k \leftarrow 0, \mu_k \leftarrow 0$

```

1 while  $\|F(u_k)\| > \epsilon$  do
2   find  $d_k$  according to (9) (with  $u_k$  instead of  $x_k$ )
3   if  $\|d_k\| \approx 0$  then
4      $x^* \leftarrow x_k$ 
5     break
6   end
7   find  $\tau_k$  according to (8) (with  $u_k$  instead of  $x_k$ )
8    $z_k \leftarrow u_k + \tau_k d_k$ 
9   if  $z_k \in \mathcal{C}$  and  $\|F(z_k)\| \leq \epsilon$  then
10     $x^* \leftarrow z_k$ 
11    break
12  end
13   $\mu_k \leftarrow \frac{F(z_k)^T (u_k - z_k)}{\|F(z_k)\|^2}$ 
14   $x_{k+1} \leftarrow P_{\mathcal{C}}(u_k - \gamma \mu_k F(z_k))$ 
15  calculate  $u_{k+1}$  by (23)
16   $k \leftarrow k + 1$ 
17   $x^* \leftarrow u_k$ 
18 end
  
```

Lemma 3.10 *The search direction d_k generated by Algorithm 2 satisfies the sufficient descent condition*

$$F(u_k)^T d_k \leq -\alpha_{\min} \|F(u_k)\|^2. \quad (28)$$

and the trust region property

$$\alpha_{\min} \|F(u_k)\| \leq \|d_k\| \leq \left(\alpha_{\max} + \frac{2}{\psi} \right) \|F(u_k)\|. \quad (29)$$

Proof We omit the proof of this Lemma as it can be easily derived following the same approach as in the proof of Lemma 3.5. $\square$

Lemma 3.11 *Suppose that Assumption (A1) and (A2) hold. If $x^* \in \Gamma$ and the sequence $\{x_k\}$ is generated by Algorithm 2. Then $\{\|x_k - x^*\|\}$ is convergent and $\{x_k\}$ is bounded.*

Proof From the line search procedure (8), it is easy to see that $\mu_k \geq 0$ (line 13 of Algorithm 2). Observe that

$$\begin{aligned} F(z_k)^T(u_k - x^*) &= F(z_k)^T(u_k - z_k + z_k - x^*) \\ &= F(z_k)^T(u_k - z_k) + F(z_k)^T(z_k - x^*). \end{aligned} \quad (30)$$

For any $x^* \in \Gamma$, $F(x^*)^T(z_k - x^*) = 0$ for all k . Thus, by (A2), $F(z_k)^T(z_k - x^*) \geq 0$. Using this fact together with the nonexpansive property of P_C , the line search (8) and (30) we can deduce that

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &= \|P_C(u_k - \gamma\mu_k F(z_k)) - P_C(x^*)\|^2 \\ &\leq \|u_k - \gamma\mu_k F(z_k) - x^*\|^2 \\ &\leq \|u_k - x^*\|^2 - 2\gamma\mu_k F(z_k)^T(u_k - x^*) + \gamma^2\mu_k^2 \|F(u_k)\|^2 \\ &= \|u_k - x^*\|^2 - 2\gamma\mu_k \langle F(z_k), u_k - z_k \rangle - 2\gamma\mu_k F(z_k)^T(z_k - x^*) \\ &\quad + \gamma^2\mu_k^2 \|F(u_k)\|^2 \\ &\leq \|u_k - x^*\|^2 - 2\gamma\mu_k F(z_k)^T(u_k - z_k) + \gamma^2\mu_k^2 \|F(u_k)\|^2 \\ &= \|u_k - x^*\|^2 - \gamma(2 - \gamma)\sigma^2 \|u_k - z_k\|^4. \end{aligned} \quad (31)$$

From the above relation, we have

$$\begin{aligned} \|x_{k+1} - x^*\| &\leq \|u_k - x^*\| \\ &\leq \|x_k - x^*\| + t_k \|x_k - x_{k-1}\|. \end{aligned} \quad (32)$$

By Lemma 2.4, $\{\|x_k - x^*\|\}$ is convergent, and hence $\{x_k\}$ is bounded. $\square$

Remark 3.12 Since the sequence $\{x_k\}$ is bounded, by virtue of (25), it can be easily seen that $\{u_k\}$ is bounded. Therefore, by the continuity of F , it follows that $\|F(u_k)\|$ is bounded. Thus, from (29), we can see that the search direction d_k generated by Algorithm 2 is also bounded.

Lemma 3.13 *Let assumptions (A1) and (A2) hold, and let $\{x_k\}$, $\{u_k\}$ and $\{z_k\}$ be generated by Algorithm 2. Then*

$$\lim_{k \rightarrow \infty} \|z_k - u_k\| = 0. \quad (33)$$

Proof From the definition of u_k , one sees that

$$\begin{aligned} \|u_k - x^*\|^2 &= \|x_k - t_k(x_k - x_{k-1}) - x^*\|^2 \\ &= \|(x_k - x^*) + t_k(x_k - x_{k-1})\|^2 \\ &= \|x_k - x^*\|^2 + 2t_k(x_k - x^*)^T(x_k - x_{k-1}) + t_k^2 \|x_k - x_{k-1}\|^2 \\ &\leq \|x_k - x^*\|^2 + 2t_k \|x_k - x^*\| \|x_k - x_{k-1}\| + t_k^2 \|x_k - x_{k-1}\|^2 \\ &= \|x_k - x^*\|^2 + t_k \|x_k - x_{k-1}\| (2\|x_k - x^*\| + \|x_k - x_{k-1}\|) \\ &\leq \|x_k - x^*\|^2 + t_k \|x_k - x_{k-1}\| m_1 \end{aligned} \quad (34)$$

for some $m_1 > 0$. Combining (31) and (34), we obtain

$$\gamma(2 - \gamma)\sigma^2 \|u_k - z_k\|^4 \leq \|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2 + m_1 t_k \|x_k - x_{k-1}\|. \quad (35)$$

Adding (35) for $k = 0, 1, 2, \dots$ and taking note of relation (26), we have

$$\begin{aligned} \gamma(2 - \gamma)\sigma^2 \sum_{k=0}^{\infty} \|u_k - z_k\|^4 &\leq \sum_{k=0}^{\infty} (\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2) \\ &\quad + m_1 \sum_{k=0}^{\infty} t_k \|x_k - x_{k-1}\| \\ &\leq \|x_0 - x^*\|^2 + m_1 \sum_{k=0}^{\infty} t_k \|x_k - x_{k-1}\| \\ &< \infty. \end{aligned} \quad (36)$$

An immediate consequence of the above relation is that $\lim_{k \rightarrow \infty} \tau_k \|d_k\| = 0$. This completes the proof. $\square$

Theorem 3.14 *Under Assumptions (A1) and (A2), the sequence $\{u_k\}$ generated by Algorithm 2 converges to a solution of (1).*

Proof Following similar argument as in the proof of Theorem 3.8, we can easily show that

$$\liminf_{k \rightarrow \infty} \|F(u_k)\| = 0. \quad (37)$$

Since $\{u_k\}$ and $\{d_k\}$ are bounded sequences without loss of generality, we assume that $u_{k_i} \rightarrow u^*$ and $d_{k_i} \rightarrow d^*$ as $i \rightarrow \infty$. By virtue of the continuity of F , we can get

$$\lim_{i \rightarrow \infty} \|F(u_{k_i})\| = \|F(u^*)\| = 0. \quad (38)$$

Thus, by (27), we can easily observe that as $i \rightarrow \infty$, $x_{k_i} \rightarrow u^*$. Hence, we can deduce that $u^* \in \mathcal{C}$ due to the clear connection that $\{x_k\} \in \mathcal{C}$ and the closedness of $\mathcal{C}$. This, in combination with (38) means that $u^* \in \Gamma$. If we substitute u^* for x^* in (32) and utilize the fact that $\lim_{i \rightarrow \infty} x_{k_i} = u^*$, we have

$$\lim_{k \rightarrow \infty} \|x_k - u^*\| = \lim_{i \rightarrow \infty} \|x_{k_i} - u^*\| = 0,$$

which further implies that the whole sequence $\{x_k\}$ converges to $u^* \in \Gamma$. $\square$

4 Numerical experiments

In this section, we perform some numerical experiments to show the computational efficiency of the proposed iterative methods. For the implementation, all algorithms were written in Julia [47] (version 1.8.5) and executions are performed on a PC equipped with an Intel Core(TM) i9-9900K processor running at 3.6 GHz and 32GB of RAM¹

4.1 Experiment 1: large-scale nonlinear systems

In this subsection, we conduct numerical experiments to demonstrate the effectiveness of **STTDFPM** and **ISTTDFPM** methods in solving large-scale nonlinear equations. We compare the performance of **STTDFPM** and **ISTTDFPM** with existing projection methods, namely, **MOPCG** [48, Algorithm 2.1], **CGDFPM** [49, Algorithm 2.1] and **AHDFPM** [50, Algorithm 1: Line Search (8)]. The comparison is carried out in terms of number of iterations, function evaluations and CPU time. To showcase the wide-ranging effectiveness of the **STTDFPM** and **ISTTDFPM** methods, we conducted numerical tests on 12 benchmark test problems. These test problems are listed in Table 1. Each problem is given a descriptive name along with the reference from which it is taken. The set $\mathcal{C}$ for each problem is listed in the last column. These sets are defined for each $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ as follows

$$\begin{aligned} \mathbb{R}_+^n &= \{x \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } 1 \leq i \leq n\} \\ (\kappa \mathbb{1} + \mathbb{R}^n)_+ &= \{x \in \mathbb{R}^n \mid x_i + \kappa \geq 0 \text{ for all } 1 \leq i \leq n\}, \quad \kappa \in \mathbb{R} \\ \mathcal{H}_-^v &= \{x \in \mathbb{R}^n \mid \sum_{i=1}^n x_i \leq v\}, \quad v \in \mathbb{R} \end{aligned}$$

All tests were carried out in three different dimensions, namely, $n = 10^3$, 10^4 and 10^5 and with 14 different initial points (39).

To enhance the effectiveness of **STTDFPM** and **ISTTDFPM**, we carefully select parameters that are best suited for their respective algorithms. The specific parameter choice for both **ISTTDFPM** and **ISTTDFPM** are

$$\begin{aligned} t = 0.11, \quad \beta = 0.5, \quad \sigma = 0.01, \quad \gamma = 1.8, \quad \alpha_{\min} = 10^{-10}, \quad \alpha_{\max} = 10^{30}, \\ r = 0.1, \quad \psi = 0.2, \quad \eta_1 = 0.001, \quad \eta_2 = 0.6 \end{aligned}$$

¹ All codes are available at <https://github.com/mmogib/STTDFPM>.

Table 1 List of test problems

Problem	Name [Reference]	$\mathcal{C}$
1	ENGVAL1Grad [52, gradient of ENGVAL1]	n
2	Exponential [53, Problem 9]	$\frac{n}{+}$
3	ExponentialSineCosine [54, Problem 1]	$\mathcal{H}_{-}^n \cap (\mathbb{1} + \frac{n}{+})_{+}$
4	Logarithmic [55, Problem 10]	$(\mathbb{1} + \frac{n}{+})_{+}$
5	ModifiedNonsmoothSine [56, Problem 2]	$\mathcal{H}_{-}^n \cap (\mathbb{1} + \frac{n}{+})_{+}$
6	ModifiedNonsmoothSine2 [54, Problem 2]	$\mathcal{H}_{-}^n \cap (\mathbb{1} + \frac{n}{+})_{+}$
7	ModifiedTridiagonal [49, Problem 4.4]	$\mathcal{H}_{-}^1 \cap \frac{n}{+}$
8	ModifiedTrig1 [49, Problem 4.2]	$(3\mathbb{1} + \frac{n}{+})_{+}$
9	NonmoothLogarithmic [48, Problem 4.2]	$\frac{n}{+}$
10	NonsmoothSine [57, Problem 1]	$\mathcal{H}_{-}^n \cap \frac{n}{+}$
11	PolynomialII [53, Problem 9]	$\frac{n}{+}$
12	SmoothSine [57, Problem 2]	$\frac{n}{+}$

The parameters for the compared methods are obtained from their respective references. All methods were implemented in Julia and tested with same initial points.

$$\left. \begin{aligned} x_0^1 &= (0, \dots, 0)^T, & x_0^2 &= (0.2, \dots, 0.2)^T, & x_0^3 &= (0.4, \dots, 0.4)^T, \\ x_0^4 &= (0.5, \dots, 0.5)^T, & x_0^5 &= (0.6, \dots, 0.6)^T, & x_0^6 &= (0.8, \dots, 0.8)^T, \\ x_0^7 &= (1, \dots, 1)^T, & x_0^8 &= (1.1, \dots, 1.1)^T, & x_0^9 &= (1 - \frac{1}{n}, 1 - \frac{1}{n}, \dots, 0)^T, \\ x_0^{10} &= (1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n})^T, & x_0^{11} &= (0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n})^T, & x_0^{12} &= (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})^T, \\ x_0^{13} &= (\frac{1}{3}, \frac{1}{9}, \dots, \frac{1}{3^n})^T, & x_0^{14} &= (\frac{1}{n}, \frac{2}{n}, \dots, 1)^T. \end{aligned} \right\} \quad (39)$$

Furthermore, all algorithms are terminated if any of the following condition is satisfied

$$(i) \|F_k\| \leq \epsilon, \quad (ii) \|d_k\| \leq \frac{\epsilon}{10} \quad \text{or} \quad (iii) k > 2000.$$

where $\epsilon = 10^{-11}$ and $\|F_k\|$ represents the norm of the value of the function at iterate k . To assess and compare the performance of the different methods, we use of the Dolan and Moré performance profiles introduced in [51]. The performance profiles of the proposed methods and compared methods are depicted in Figs. 1, 2, and 3. In the figures, $\rho(\tau)$ denotes the probability that the solver will outperform the other solvers. From Figs. 1, 2, and 3, it can be observed that **STTDFPM** method achieved approximately 50%, 61%, and 45% success rates in terms of the number of iterations, the number of function evaluations, and the CPU time respectively. **ISTTDFPM**, although outperforms both **MOPCG** and **CGDFPM**, surprisingly performs poorly

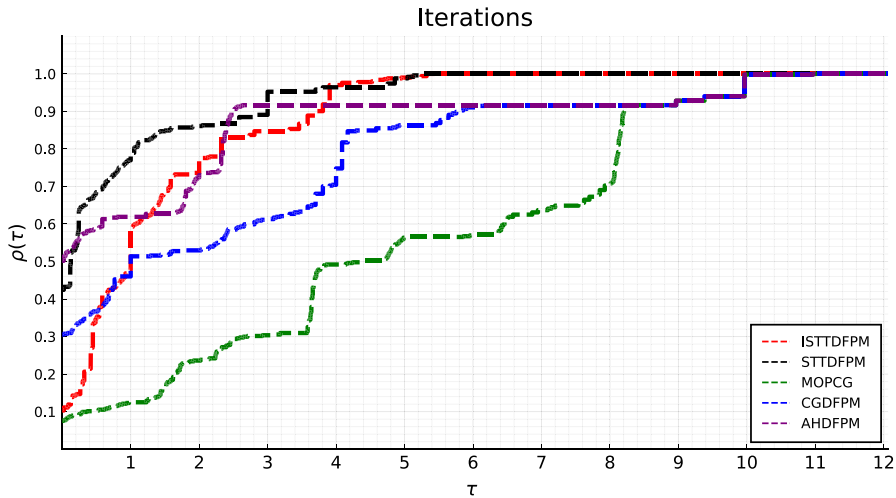


Fig. 1 Performance profiles of iterations

compared to **STTDFPM** and **AHDFPM** in all three categories. For detailed numerical experiments, readers can refer to the following link.²

4.2 Experiment 2: signal reconstruction

In this subsection, we present an application of our method in the field of compressed sensing. Compressed sensing is a branch of signal processing that focuses on reconstructing a sparse signal from measured data. Specifically, our objective in this context is to recover a sparse signal $x \in \mathbb{R}^n$ using the following observation model

$$b = Ax + c \quad (40)$$

where $b \in \mathbb{R}^m$ is the observed data, $A \in \mathbb{R}^{m \times n}$ ($m \ll n$), and $c \in \mathbb{R}^m$ is a Gaussian noise. Due to the undetermined and noisy nature of the system (40), regularization techniques are employed to reconstruct the sparse signal x . In our numerical simulations, we considered the following unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x) = \phi \|x\|_1 + \frac{1}{2} \|Ax - b\|_2^2, \quad (41)$$

where $\|\cdot\|_1$ is the ℓ_1 -norm and $\phi > 0$ is a constant to balance the data fitting and sparsity. The unconstrained optimization problem (41), is a reformulation technique for solving (40) via the ℓ_1 -regularization technique. It has been shown in [58] that solving the minimization problem (41) is equivalent to solving (1) with $F : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ defined as

$$F(z) = \min \{z, Hz + c\}, \quad (42)$$

² <https://github.com/mmogib/STTDFPM>.

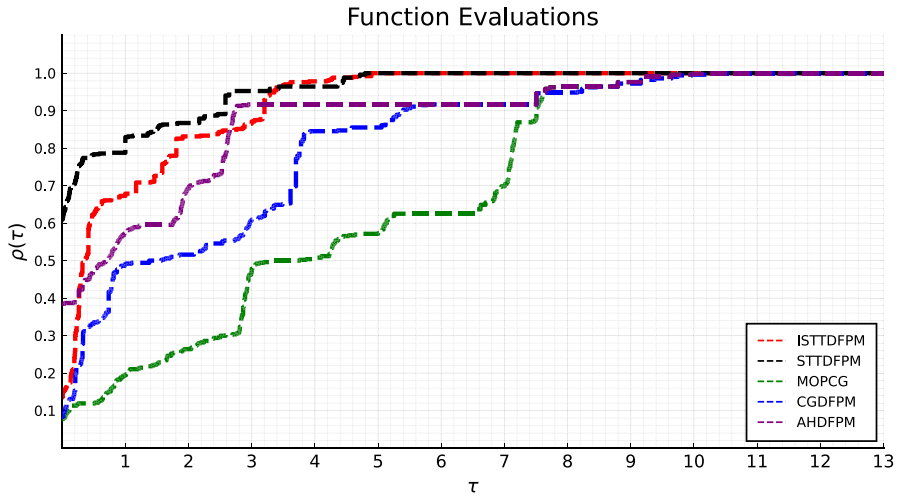


Fig. 2 Performance profiles of function evaluations

where

$$z = \begin{bmatrix} \max\{x, 0\} \\ \max\{-x, 0\} \end{bmatrix}, \quad c = \begin{bmatrix} \phi \mathbb{1} - A^T b \\ \phi \mathbb{1} + A^T b \end{bmatrix}, \quad H = \begin{bmatrix} A^T A & -A^T A \\ -A^T A & A^T A \end{bmatrix}.$$

Here, $\mathbb{1}$ is a vector where all of its components equal 1. The function F is clearly continuous and monotone [59, Lemma 2.2], therefore pseudomonotone. Thus, (42) can be solved by all algorithms (Fig. 4).

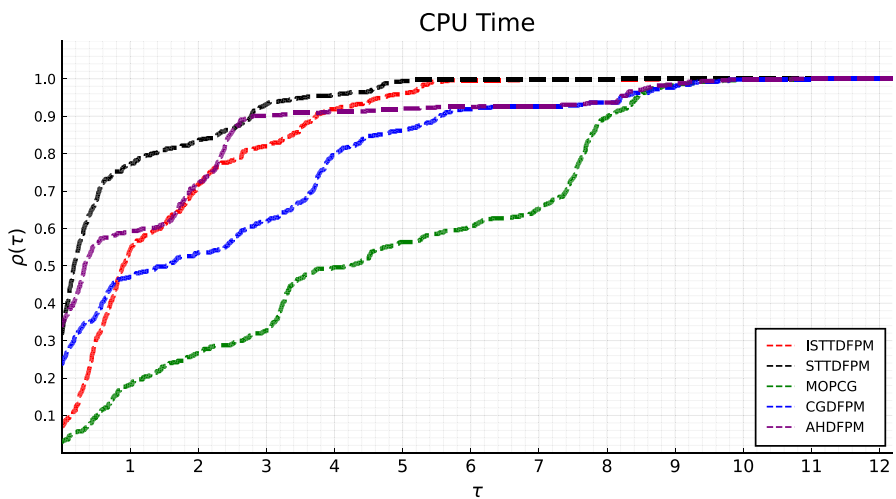


Fig. 3 Performance profiles of CPU time

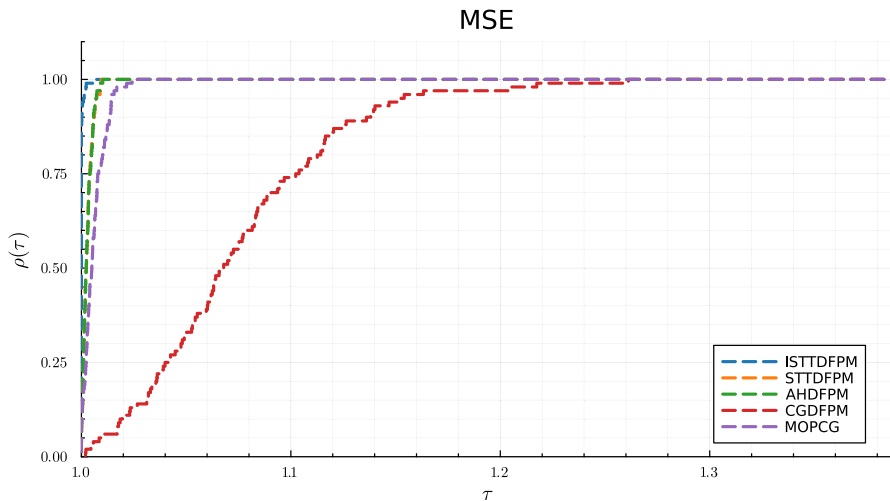


Fig. 4 Performance profile of CS MSE

In the numerical implementation, the matrix A is randomly generated and the observed data is computed using a Gaussian noise whose elements are generated from shape i.i.d. normal distributions $\mathcal{N}(0, 0.01)$. The iterative process starts with the measurement signal, that is, $x_0 = A^T b$. We select $n = 2^{11}$ and $m = 2^9$ for (40). The original signal x_0 contains 2^7 nonzero elements randomly. The experiment continues until the following condition is satisfied

$$\frac{|f(x_k) - f(x_{k-1})|}{|f(x_{k-1})|} < 10^{-4} \quad (43)$$

To assess the performance of **STDFPM** in reconstructing sparse signals, we compare it against three other methods: **AHDFPM** [50], **CGDFPM** [49], and **MOPCG** [48]. This comparison allows us to evaluate how well **STDFPM** performs relative to these methods. To quantitatively evaluate the quality of the restored signal, we utilize the Mean Squared Error (MSE) metric. A smaller MSE value signifies a higher quality of signal restoration. To ensure a fair comparison, we conducted 100 experiments.

Table 2 Average MSE of 100 random signals with ($n = 2^{11}$, $m = 2^9$ and $r = 2^7$)

Algorithm	Average MSE
ISTDFPM	4.40350E-06
STDFPM	4.41477E-06
AHDFPM	4.41486E-06
MOPCG	4.42780E-06
CGDFPM	4.71106E-06

The comprehensive results of these experiments are presented in Table 2.³ The table clearly demonstrates that our algorithm performs better in reconstructing the sparse signal compared to other methods.

5 Conclusion

In this paper, systems of nonlinear equations are considered. A spectral three-term conjugate gradient method framework for unconstrained optimization is extended, and two derivative-free methods **STTDFPM** and **ISTTDFPM** are proposed. The proposed method effectively leverages the advantages of both the conjugate gradient method and the projection method. It exhibits a descent search direction, satisfies the trust region property, and achieves global convergence under mild conditions. The numerical results obtained further validate the competitiveness of our method in efficiently solving large-scale problems. Although, **ISTTDFPM** is an accelerated version of **STTDFPM** using the inertia extrapolation technique, we did not obtain significant improvement as we expected. This in turn implies that the inertial technique may not always accelerate the convergence of derivative-free iterative methods for solving nonlinear equations. This should be for future investigation.

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Code availability <https://github.com/mmogib/STTDFPM>

Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors declare no competing interests.

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³ <https://github.com/mmogib/STTDFPM>.

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